

# TOBY LAU

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## EDUCATION

### University of British Columbia

B.Sc. Computer Science – GPA: 3.95

May 2028 (Expected)

Vancouver, BC

## EXPERIENCE

### Squarepoint Capital

Software Engineer Intern – Data Services (Commodities)

May 2026 – Aug 2026

Montreal, QC

- Created a monitoring platform for alpha researchers to aggregate and visualize health check results across **1100+** alphas.
- Cut p95 API latency by **22%** by merging an Apollo subgraph with downstream gRPC services across **35+** prod endpoints.
- Built a polling scheduler and worker pipeline to ingest and process **~ 20k** alpha metrics/second from a time-series database.
- Implemented cron-derived lookback window calculation and status detection to detect stale metrics across **100+** strategies.
- Designed a YAML-database sync mechanism enabling **50+** desks to version-control dashboard configurations through GitLab.

### Planview

Software Engineer Intern – Planview Copilot

Jan 2025 – Aug 2025

Vancouver, BC

- Reduced chat response latency by **50%+** by implementing concurrent tool calls to AWS Lambdas in Java-based chat service.
- Unblocked a major product release for **2k+** users by implementing a dual-token handler during an auth service migration.
- Implemented a log obfuscator/forwarder that routed **21k+** sanitized daily logs to Datadog, retaining raw logs in CloudWatch.
- Delivered a self-service RAG platform for **100+** tenants with S3 and Bedrock, enabling document uploads and analysis/Q&A.
- Built a hybrid tool-calling/MCP chat service and registry to support existing tools while integrating newer MCP servers.

### UBC Department of Computer Science

Undergraduate Teaching Assistant – CPSC 210 (Introduction to Object Oriented Programming)

Sep 2024 – Dec 2024

Vancouver, BC

- Taught OOP/OOD, design patterns, and TDD with JUnit to **300+** students; mentored **25+** students in weekly labs/office hours.

### APL Zuellig Pharma

Software Engineer Intern – Distribution Resource Planning

Oct 2021 – Jan 2022

Remote, Indonesia

- Replaced a complex Excel workflow for stock/distribution planning with a Python GUI across **46k+** SKUs in **25** warehouses.

## EXTRACURRICULARS

### Databricks

Student Fellow

June 2026 – Present

Remote

- **1 of 36** selected from **5k+** globally to represent Databricks campus/user groups, with mentorship from Databricks engineers.

## PROJECTS

### Extreme Vocals Classifier | Python, PyTorch, Numpy, Scipy, Librosa, Google Colab

- Synthesized a **500+** extreme vocal sample dataset by isolating vocals with Demucs, extracting and processing relevant vocal regions with Numpy/Scipy (RMS normalization, high-pass filter), and augmenting vocals with librosa from **57+** songs.

### 2026 FIFA World Cup Match Report ETL Pipeline | Python, Pandas, Spark, Databricks

- Built a Spark/Databricks pipeline to extract and structure data from **32** FIFA match report PDFs. Combined pdfplumber text parsing with image-hash-based icon classification to write normalized match, team, and player tables to Delta Lake.

### Membership Portal | React, TypeScript, PostgreSQL, Stripe, Supabase, Google Cloud Platform

- Supported **250+ MAU** and **\$5k+** growth in revenue by building payments, registration flows, and admin tools from **0→1**.

### Serverless Link Tracker | Python, Google Cloud Functions, Firestore

- Built a serverless link redirector and companion CLI to track clicks with per-link metadata and async webhook notifications.

## TECHNICAL SKILLS

**Languages:** Java, Python, TypeScript, JavaScript, HTML, CSS, SQL, C, C++

**Frameworks:** React, Spring (Boot), Next.js, Express, Flask, FastAPI, JUnit, Jest

**Technologies:** AWS, GCP, Supabase, PostgreSQL, Docker, Kubernetes, ArgoCD, Liquibase/Alembic, GraphQL, gRPC

**Tools:** Git, Postman/Insomnia, Datadog, Elastic Search, Jupyter, DBeaver, Vim, Figma, Jira, IntelliJ, Claude Code